



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



SILVER OAK UNIVERSITY

Silver Oak College of Computer Application

Course: Foundation of Mathematics | Course Code: MDC5001C | Semester: 1

List of Tutorials

UNIT 3: OPERATIONS RESEARCH AND LINEAR PROGRAMMING (CO Mapping: CO3)

Tutorial 1: Foundations of OR — Basic Concepts, Scope & Applications

Topics covered: Basic concepts, scope and applications of Operations Research

Q.No	Question	Difficulty	Bloom's Level	CO
1	Define Operations Research and state its main characteristics.	Easy	Remember	CO3
2	List and briefly explain any four fields where Operations Research is applied.	Easy	Understand	CO3
3	Explain the phases involved in solving a real-world problem using the OR approach.	Moderate	Understand	CO3
4	A manufacturing company wants to decide how many units of two products to produce with limited raw material. Explain how OR can help structure this decision.	Moderate	Apply	CO3
5	Discuss the scope and limitations of Operations Research in decision-making.	Challenging	Analyze	CO3

Tutorial 2: Visualizing Solutions — Graphical Method for Two-Variable LPP

Topics covered: Graphical solution for two-variable LPP; feasible region, optimal solution

Q.No	Question	Difficulty	Bloom's Level	CO
6	Define feasible region and optimal solution in the context of an LPP.	Easy	Remember	CO3
7	State the steps involved in solving a two-variable LPP graphically.	Easy	Understand	CO3
8	Maximize $Z = 3x + 5y$ subject to $x + 2y \leq 14$, $3x - y \geq 0$, $x - y \leq 2$, $x, y \geq 0$, using the graphical method.	Moderate	Apply	CO3
9	Minimize $Z = 20x + 10y$ subject to $x + 2y \leq 40$, $3x + y \geq 30$, $4x + 3y \geq 60$, $x, y \geq 0$, using the graphical method.	Moderate	Apply	CO3
10	Solve graphically: Maximize $Z = 5x + 4y$ subject to $6x + 4y \leq 24$, $x + 2y \leq 6$, $x, y \geq 0$.	Moderate	Analyze	CO3
11	Explain how the corner-point (extreme point) theorem is used to identify the optimal solution in the graphical method.	Challenging	Analyze	CO3
12	A furniture company makes chairs and tables with limited wood and labor. Formulate and solve the LPP graphically to maximize profit.	Challenging	Evaluate	CO3

Tutorial 3: Special Cases — Multiple, Unbounded & Infeasible Solutions

Topics covered: Multiple/unbounded/infeasible cases

Q.No	Question	Difficulty	Bloom's Level	CO
13	Define multiple optimal solutions, unbounded solutions, and infeasible solutions in an LPP.	Easy	Remember	CO3
14	Explain, with a graphical illustration, how a multiple optimal solution arises when the objective function line is parallel to a constraint.	Easy	Understand	CO3
15	Solve graphically: Maximize $Z = 2x + 4y$ subject to $x + 2y \leq 8$, $x, y \geq 0$, and identify whether the solution is unique or has multiple optima.	Moderate	Apply	CO3
16	Show that Maximize $Z = x + y$ subject to $x - y \leq 1$, $-x + y \leq 2$, $x, y \geq 0$ has an unbounded solution.	Moderate	Analyze	CO3
17	Show that the constraints $x + y \leq 2$, $x + y \geq 5$, $x, y \geq 0$ result in an infeasible LPP.	Moderate	Analyze	CO3

Q.No	Question	Difficulty	Bloom's Level	CO
18	Discuss the practical implications of an unbounded or infeasible LPP for a real business planning problem.	Challenging	Evaluate	CO3

Tutorial 4: Standardizing Problems — Standard Form of LPP

Topics covered: Standard form of LPP

Q.No	Question	Difficulty	Bloom's Level	CO
19	Define the standard form of a Linear Programming Problem.	Easy	Remember	CO3
20	Explain the role of slack, surplus, and artificial variables in converting an LPP to standard form.	Easy	Understand	CO3
21	Convert the LPP: Maximize $Z = 4x + 3y$ subject to $2x + y \leq 10$, $x + 3y \leq 15$, $x, y \geq 0$, into standard form.	Moderate	Apply	CO3
22	Convert the LPP: Minimize $Z = 5x + 2y$ subject to $2x + y \geq 8$, $x + 3y \geq 9$, $x, y \geq 0$, into standard form using surplus and artificial variables.	Moderate	Apply	CO3
23	Explain why artificial variables are penalized in the objective function (Big-M method context), with an example.	Challenging	Analyze	CO3

Tutorial 5: Tableau Method — Introduction to Simplex Algorithm

Topics covered: Introduction to Simplex algorithm and Tableau method

Q.No	Question	Difficulty	Bloom's Level	CO
24	State the basic steps of the Simplex algorithm.	Easy	Remember	CO3
25	Explain the terms "basic variable," "non-basic variable," and "pivot element" in the Simplex Tableau.	Easy	Understand	CO3
26	Solve using the Simplex method: Maximize $Z = 3x + 2y$ subject to $x + y \leq 4$, $x + 3y \leq 6$, $x, y \geq 0$.	Moderate	Apply	CO3
27	Solve using the Simplex Tableau: Maximize $Z = 5x + 4y$ subject to $6x + 4y \leq 24$, $x + 2y \leq 6$, $x, y \geq 0$.	Moderate	Apply	CO3
28	Explain the criterion used in the Simplex method to select the entering and leaving variables at each iteration.	Moderate	Analyze	CO3

Q.No	Question	Difficulty	Bloom's Level	CO
29	Solve using Simplex: Maximize $Z = 40x + 30y$ subject to $2x + y \leq 40$, $x + y \leq 25$, $x, y \geq 0$, showing all tableau iterations.	Challenging	Analyze	CO3
30	Explain how the Simplex method detects an unbounded solution during the tableau iterations.	Challenging	Evaluate	CO3
31	Discuss why the Simplex method is generally more efficient than the graphical method for solving LPPs with more than two variables.	Challenging	Evaluate	CO3

Tutorial 6: The Mirror Problem — Duality & Dual Formulation

Topics covered: Concept of Duality, formulation of Dual Problem

Q.No	Question	Difficulty	Bloom's Level	CO
32	Define the concept of duality in Linear Programming.	Easy	Remember	CO3
33	State the general rules for converting a primal LPP into its dual.	Easy	Understand	CO3
34	Write the dual of: Maximize $Z = 3x + 5y$ subject to $x + 4y \leq 24$, $3x + y \leq 21$, $x, y \geq 0$.	Moderate	Apply	CO3
35	Find the dual of: Minimize $Z = 8x + 6y$ subject to $4x + 2y \geq 60$, $2x + 4y \geq 48$, $x, y \geq 0$.	Moderate	Apply	CO3
36	Explain the economic interpretation of the dual variables (shadow prices) in a resource allocation problem.	Moderate	Analyze	CO3
37	State and explain the weak duality theorem and its significance in verifying an optimal LPP solution.	Challenging	Evaluate	CO3

Tutorial 7: Initial Solutions — NWC Rule & Least Cost Method

Topics covered: Transportation Problem — Initial Basic Feasible Solution using North-West Corner Rule and Least Cost Method

Q.No	Question	Difficulty	Bloom's Level	CO
38	Define a Transportation Problem, and state the condition for it to be balanced.	Easy	Remember	CO3
39	Explain the steps of the North-West Corner (NWC) Rule for finding an initial basic feasible solution.	Easy	Understand	CO3

Q.No	Question	Difficulty	Bloom's Level	CO
40	A company has 3 factories supplying 3 warehouses with given supply/demand and cost matrix. Find the IBFS using the NWC Rule.	Moderate	Apply	CO3
41	For the same transportation problem in Q40, find the IBFS using the Least Cost Method.	Moderate	Apply	CO3
42	Compare the North-West Corner Rule and the Least Cost Method in terms of solution quality and total transportation cost.	Moderate	Analyze	CO3
43	Explain why the total supply must equal total demand for a transportation problem to be considered balanced, and how an unbalanced problem is handled.	Challenging	Analyze	CO3
44	A transportation problem has 3 sources and 4 destinations. Formulate the cost matrix and find the IBFS using the Least Cost Method, verifying the total cost.	Challenging	Evaluate	CO3

Tutorial 8: Optimizing Transport — MODI Method & Degeneracy in TP

Topics covered: Optimality test using MODI method; Degeneracy in Transportation Problems

Q.No	Question	Difficulty	Bloom's Level	CO
45	Define the MODI (Modified Distribution) method and state its purpose.	Easy	Remember	CO3
46	Explain what degeneracy means in a transportation problem and when it occurs.	Easy	Understand	CO3
47	Starting from an IBFS obtained by the Least Cost Method, apply the MODI method to test for optimality and find any improved solution.	Moderate	Apply	CO3
48	Compute the u_i and v_j values for a given transportation tableau and determine the opportunity cost for each unoccupied cell.	Moderate	Apply	CO3
49	Explain how a degenerate basic feasible solution is resolved using an epsilon (ϵ) allocation.	Moderate	Analyze	CO3
50	For a transportation problem with 4 sources and 3 destinations, apply the MODI method fully to reach the optimal solution, addressing any degeneracy encountered.	Challenging	Evaluate	CO3

Q.No	Question	Difficulty	Bloom's Level	CO
51	Discuss why degeneracy can cause cycling in the MODI method, and how this is typically avoided in practice.	Challenging	Evaluate	CO3

Tutorial 9: One-to-One Matching — Assignment Problem (Hungarian Method)

Topics covered: Assignment Problem — Hungarian Method

Q.No	Question	Difficulty	Bloom's Level	CO
52	Define the Assignment Problem and explain how it differs from the Transportation Problem.	Easy	Remember	CO3
53	State the steps of the Hungarian Method for solving an assignment problem.	Easy	Understand	CO3
54	Solve a 3×3 assignment problem (given a cost matrix of 3 workers and 3 jobs) using the Hungarian Method to minimize total cost.	Moderate	Apply	CO3
55	Solve a 4×4 assignment problem using the Hungarian Method, given a cost matrix, and find the minimum total assignment cost.	Moderate	Apply	CO3
56	Explain how row reduction and column reduction in the Hungarian Method help identify a zero-cost assignment.	Moderate	Analyze	CO3
57	Discuss how an unbalanced assignment problem (unequal number of workers and jobs) is handled using dummy rows/columns.	Challenging	Analyze	CO3
58	A company must assign 4 salespersons to 4 territories to maximize total sales. Convert this maximization problem to a minimization form and solve using the Hungarian Method.	Challenging	Evaluate	CO3

Tutorial 10: Strategic Choices — Game Theory: Saddle Point & Strategies

Topics covered: Introduction to Game Theory — Two-person zero-sum games, saddle point, pure and mixed strategies

Q.No	Question	Difficulty	Bloom's Level	CO
59	Define a two-person zero-sum game, with a simple example payoff matrix.	Easy	Remember	CO3

Q.No	Question	Difficulty	Bloom's Level	CO
60	Define saddle point and explain its significance in solving a game with pure strategies.	Easy	Understand	CO3
61	Find the saddle point (if it exists) for the payoff matrix $[[2,-1],[0,3]]$ and determine the value of the game.	Moderate	Apply	CO3
62	Determine whether the game with payoff matrix $[[1,3,-1],[0,2,4],[-2,1,3]]$ has a saddle point.	Moderate	Apply	CO3
63	For a 2×2 game with no saddle point, explain how mixed strategies and their probabilities are determined algebraically.	Moderate	Analyze	CO3
64	Solve a 2×2 game with payoff matrix $[[4,1],[2,3]]$ (no saddle point) for optimal mixed strategies and the value of the game.	Challenging	Evaluate	CO3

Tutorial 11: Simplifying Games — Dominance Principle & Graphical Method

Topics covered: Dominance principle, graphical method for games; Real-life applications

Q.No	Question	Difficulty	Bloom's Level	CO
65	State the Dominance Principle used to simplify a game's payoff matrix.	Easy	Remember	CO3
66	Explain how a row or column is eliminated from a payoff matrix using dominance rules.	Easy	Understand	CO3
67	Reduce the payoff matrix $[[3,2,4],[1,4,0],[2,3,1]]$ using the dominance principle before solving.	Moderate	Apply	CO3
68	Solve a $2 \times n$ game using the graphical method, given a payoff matrix with Player A having 2 strategies.	Moderate	Apply	CO3
69	Explain how the graphical method for games relates to plotting expected payoff lines and identifying the lower/upper envelope.	Moderate	Analyze	CO3
70	Discuss a real-life application of Game Theory in business competition (e.g., pricing strategy between two firms), modeling it as a payoff matrix.	Challenging	Evaluate	CO3